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#include <mpi.h>
#include <stdlib.h>
#include <stdio.h>

#define NPROCS 8

main(int argc, char *argv[])
{
    int    rank, new_rank, sendbuf, recvbuf, size,
           ranks1[4]={0,1,2,3},ranks2[4]={4,5,6,7};

    MPI_Group orig_group,new_group;
    MPI_Comm new_comm;

    MPI_Init(&argc,&argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    if (size != NPROCS)
    {
        printf("Must specify MP_PROCS= %d. Terminating.\n",NPROCS);
        MPI_Finalize();
        exit(0);
    }

    printf("\n Enter value for process%d: ",rank);
    scanf("%d",&sendbuf);

    /* Extract the original group handle */
    MPI_Comm_group(MPI_COMM_WORLD, &orig_group);

    /* Divide tasks into two distinct groups based upon rank */
    if (rank < NPROCS/2)
    {
        MPI_Group_incl(orig_group, NPROCS/2, ranks1, &new_group);
    }
    else
    {
        MPI_Group_incl(orig_group, NPROCS/2, ranks2, &new_group);
    }

    /* Create new new communicator and then perform collective communications */
    MPI_Comm_create(MPI_COMM_WORLD, new_group, &new_comm);

    if (rank < 3)
    {
        MPI_Allreduce(&sendbuf, &recvbuf, 1, MPI_INT, MPI_SUM, new_comm);
    }
}

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else
{
MPI_Allreduce(&sendbuf, &recvbuf, 1, MPI_INT, MPI_MAX, new_comm);
}

MPI_Group_rank(new_group,&new_rank);

printf("rank= %d newrank= %d sendbuf= %d recvbuf= %d\n"
,rank,new_rank,sendbuf,recvbuf);

MPI_Finalize();
}

```

Output

```

mpicc group.c -o grp
mpirun -np 8 grp

```

```

Enter value for process0: Enter value for process1:
Enter value for process4: Enter value for process5:
Enter value for process2: Enter value for process6:
Enter value for process3: Enter value for process7: 0
10
40
50
20
60
30
70
rank= 0 newrank= 0 sendbuf= 0 recvbuf= 60
rank= 2 newrank= 2 sendbuf= 20 recvbuf= 60
rank= 4 newrank= 0 sendbuf= 40 recvbuf= 70
rank= 6 newrank= 2 sendbuf= 60 recvbuf= 70
rank= 3 newrank= 3 sendbuf= 30 recvbuf= 60
rank= 7 newrank= 3 sendbuf= 70 recvbuf= 70
rank= 5 newrank= 1 sendbuf= 50 recvbuf= 70
rank= 1 newrank= 1 sendbuf= 10 recvbuf= 60

```